

SUPPLEMENTARY RESULTS: HMAMBA

This document contains the model-specific component and loss-function ablations for HMamba, accompanying *Beyond the Majority Bin: A Methodological Framework for Balanced Automatic Pronunciation Assessment*. The cross-model comparisons and experimental reproducibility settings are reported in the paper’s supplementary material.

I. READING GUIDE

Baseline denotes the original model and training procedure. The +CV configuration introduces speaker-independent grouped cross-validation using `GroupKFold(n_splits=5)` without shuffling. Both +DB and +MS build on CV: +DB adds Dynamic Balanced MSE, whereas +MS adds Macro-MSE-based checkpoint selection. Complete BOMA combines CV, DB, and MS. In the loss-function ablations, CV and MS are held fixed while conventional MSE, DB-MSE, and Score-Balanced Loss (SB) are compared.

CCC-BS measures distribution-balanced agreement and PCC measures linear association; higher values are preferred for both. Micro-MSE and Macro-MSE are error measures on the unified $[0, 10]$ score scale, so lower values are preferred. Boldface marks the best value in each comparison. These metrics express complementary evaluation objectives; improvements under balanced criteria need not coincide with improvements under sample-weighted criteria.

II. RESULTS

For HMamba, BOMA is best or tied for best on seven CCC-BS aspects, whereas CV+MS remains strongest for utterance fluency and prosody (Table S1). The Macro-MSE pattern is closely aligned: BOMA is lowest on seven aspects and CV+MS on the same two utterance dimensions (Table S2). PCC mainly favors the Baseline, except for word stress and utterance completeness (Table S3). In the loss-function ablation, DB-MSE is best on seven CCC-BS aspects and seven Macro-MSE aspects, while conventional MSE retains the best results for utterance fluency and prosody (Tables S4 and S5).

TABLE S1
ABLATION STUDY FOR HMAMBA. CORRELATIONS ARE MEASURED USING CCC-BS.

Configuration	Phone Score	Word Score (CCC-BS)			Utterance Score (CCC-BS)				
	Phoneme ↑	Acc. ↑	Stress ↑	Total ↑	Acc. ↑	Comp. ↑	Fluency ↑	Prosody ↑	Total ↑
Baseline	0.552	0.583	0.139	0.584	0.596	0.037	0.703	0.714	0.754
+ CV	0.524	0.562	0.130	0.570	0.609	0.032	0.714	0.727	0.782
+ DB	0.647	0.687	0.636	0.664	0.633	0.092	0.652	0.676	0.770
+ MS	0.542	0.585	0.150	0.603	0.607	0.033	0.744	0.760	0.765
BOMA	0.664	0.701	0.662	0.672	0.633	0.092	0.699	0.705	0.788

^a The +DB and +MS configurations are evaluated on top of the CV configuration; BOMA combines CV, DB, and MS.

^b Bold values indicate the highest CCC-BS value for each aspect within this ablation.

TABLE S2
MICRO-MSE AND MACRO-MSE ABLATION STUDY FOR HMAMBA. LOWER VALUES INDICATE BETTER PERFORMANCE.

	Phone		Word Score						Utterance Score									
	Phoneme		Acc.		Stress		Total		Acc.		Comp.		Fluency		Prosody		Total	
Configuration	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓
Baseline	1.552	13.002	1.626	11.237	0.173	9.737	1.229	9.131	0.843	2.334	0.251	27.295	0.593	5.189	0.619	4.637	0.776	3.905
+ CV	1.679	13.790	1.739	11.450	0.173	9.760	1.315	9.481	0.883	2.392	0.257	27.608	0.645	5.277	0.653	4.531	0.807	3.536
+ DB	2.778	7.965	2.982	6.170	1.098	3.475	2.466	5.764	0.967	2.068	0.255	24.853	0.640	6.007	0.680	5.130	0.867	3.520
+ MS	1.921	12.783	1.914	10.033	0.185	9.563	1.433	8.599	0.983	2.313	0.301	28.367	0.687	4.454	0.719	3.808	0.904	3.583
BOMA	4.760	6.472	5.478	5.012	1.787	3.029	3.994	4.903	1.373	1.883	0.309	23.056	0.824	5.193	0.753	4.883	1.142	3.177

^a Micro-MSE (Mic.) and Macro-MSE (Mac.) are reported on the unified $[0, 10]$ scale; lower values (↓) indicate better performance.

^b The +DB and +MS configurations are evaluated on top of CV; BOMA combines CV, DB, and MS.

^c Bold values indicate the lowest error for each metric within the model-specific ablation.

TABLE S3
PCC ABLATION STUDY FOR HMAMBA. HIGHER VALUES INDICATE BETTER PERFORMANCE.

Configuration	Phone Score	Word Score (PCC)			Utterance Score (PCC)				
	Phoneme ↑	Acc. ↑	Stress ↑	Total ↑	Acc. ↑	Comp. ↑	Fluency ↑	Prosody ↑	Total ↑
Baseline	0.739	0.708	0.366	0.718	0.807	0.278	0.848	0.843	0.829
+ CV	0.713	0.685	0.375	0.695	0.796	0.243	0.844	0.837	0.818
+ DB	0.648	0.610	0.252	0.612	0.780	0.286	0.828	0.821	0.805
+ MS	0.679	0.656	0.345	0.667	0.785	0.293	0.839	0.831	0.809
BOMA	0.580	0.546	0.220	0.558	0.761	0.272	0.822	0.819	0.788

^a The +DB and +MS configurations are evaluated on top of CV; BOMA combines CV, DB, and MS.

^b Bold values indicate the highest PCC value for each aspect within the model-specific ablation.

TABLE S4
LOSS-FUNCTION ABLATION IN TERMS OF CCC-BS FOR HMAMBA. ALL CONFIGURATIONS USE THE SAME CROSS-VALIDATION AND MACRO-MSE-BASED MODEL-SELECTION PROTOCOL; ONLY THE TRAINING LOSS IS CHANGED.

Loss	Phone Score	Word Score (CCC-BS)			Utterance Score (CCC-BS)				
	Phoneme ↑	Acc. ↑	Stress ↑	Total ↑	Acc. ↑	Comp. ↑	Fluency ↑	Prosody ↑	Total ↑
MSE	0.542	0.585	0.150	0.603	0.607	0.033	0.744	0.760	0.765
DB-MSE	0.664	0.701	0.662	0.672	0.633	0.092	0.699	0.705	0.788
SB	0.587	0.621	0.334	0.604	0.611	0.069	0.669	0.688	0.763

^a MSE corresponds to the CV+MS configuration, i.e., the BOMA validation and model-selection protocol trained with conventional MSE.

^b DB-MSE corresponds to the proposed Dynamic Balanced MSE used in the complete BOMA framework.

^c SB denotes Score-Balanced Loss, which replaces DB-MSE while keeping the remaining protocol unchanged.

^d Bold values indicate the highest CCC-BS for each aspect within the model-specific loss ablation.

TABLE S5
LOSS-FUNCTION ABLATION IN TERMS OF MICRO-MSE AND MACRO-MSE FOR HMAMBA. ALL CONFIGURATIONS USE THE SAME CROSS-VALIDATION AND MACRO-MSE-BASED MODEL-SELECTION PROTOCOL; ONLY THE TRAINING LOSS IS CHANGED.

Loss	Phone		Word Score						Utterance Score									
	Phoneme		Acc.		Stress		Total		Acc.		Comp.		Fluency		Prosody		Total	
	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓	Mic. ↓	Mac. ↓
MSE	1.921	12.783	1.914	10.033	0.185	9.563	1.433	8.599	0.983	2.313	0.301	28.367	0.687	4.454	0.719	3.808	0.904	3.583
DB-MSE	4.760	6.472	5.478	5.012	1.787	3.029	3.994	4.903	1.373	1.883	0.309	23.056	0.824	5.193	0.753	4.883	1.142	3.177
SB	2.278	11.062	2.393	9.334	0.533	6.810	1.909	7.723	0.966	2.472	0.251	26.007	0.673	5.797	0.699	5.284	0.913	4.016

^a MSE corresponds to the CV+MS configuration, i.e., the BOMA validation and model-selection protocol trained with conventional MSE.

^b DB-MSE corresponds to the proposed Dynamic Balanced MSE used in the complete BOMA framework.

^c SB denotes Score-Balanced Loss, which replaces DB-MSE while keeping the remaining protocol unchanged.

^d Micro-MSE (Mic.) and Macro-MSE (Mac.) are reported on the unified [0, 10] scale; lower values (↓) indicate better performance.

^e Bold values indicate the lowest error for each metric within the model-specific loss ablation.